

2(a)(i)	<i>any 3 of:</i> 1. (same) volume / 10 cm³, of semen ; 2. (same) speed of spinning / spun at 1275 revolutions per minute ; 3. (same) time for spinning / spun for 20 minutes; 4. (same) volume / 15 cm³, of buffer ;	3
2(a)(ii)	<i>any 1 of:</i> <i>idea that:</i> some sperm cells may get damaged ; separation of X-bearing and Y-bearing sperm is not achieved / all sperm collect at the bottom of the tube ; AVP ;	1
2(b)	<u>14</u> ;	1

Question	Answer	Marks
2(c)	<i>any 5 of:</i> <i>estimate of numbers:</i> - idea of find the volume of the sample or (volume of sample is) – (for mm³) $0.25 \times 0.25 \times 0.1 / 0.00625 / 6.25 \times 10^{-3}$ (for cm³) $0.025 \times 0.025 \times 0.01 / 0.00000625 / 6.25 \times 10^{-6}$; - divide the number of sperm by the volume or divide 1 mm³ by the volume and multiply by the number of sperm counted ; - evidence of correct method of converting to cm³ ; <i>deciding which sperm to include:</i> <i>idea of:</i> count those where sperm (fully) inside, the square / 0.0625 (mm²) / all (small) squares / haemocytometer / quadrant / grid / boxes ; <i>idea of:</i> including those in or partially in the inner double lines ; <i>idea of:</i> including those (in edge region) top and left / up and right or exclude those in edge region) bottom and right / down and left ; - AVP <i>detail of method:</i> e.g. ref. to, dilution / stirring / even spreading / immobilising sperm / use of coverslip / screenshot for counting / <i>idea of</i> only counting non- deformed / sperm ;	5